

Geography Ch-17 (Must-Do): The Savanna or Sudan Climate (Aw)

Tier 1: Must-Do | Source: GC Leong + Majid Husain (Indian & World Geography)
Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

The Savanna / Sudan Climate (Aw): Tropical Grassland with Distinct Wet & Dry Seasons

GS-I (Physical Geography - Climatology / Biogeography) Geography

■ NEWS TRIGGER

Foundation concept (GC Leong Ch-17). The Savanna or 'Sudan' climate is the TRANSITIONAL type between the equatorial forests and the hot deserts, found within the tropics (roughly 5-20 deg N & S), best developed in the Sudan belt of Africa.

■ INTRO & ORIGIN

The savanna climate has two sharply contrasted seasons: a hot rainy summer (when the ITCZ/overhead sun brings convectional rain) and a warm dry winter (when dry trade winds prevail). This wet-dry rhythm produces the classic tropical grassland - tall grass with scattered trees - the 'parkland' landscape.

Transitional between equatorial (constant rain) and desert (constant dryness): as the sun/ITCZ migrates, the savanna gets equatorial-type rain in the high-sun (summer) season and trade-wind aridity in the low-sun (winter) season. Rain concentrated in summer; total ~ 75-150 cm.

■ KEY DATA

Feature	Detail	Note
Location	5-20 deg N & S, within tropics	Sudan (Africa), Llanos, Campo Cerrado, N. Australia
Seasons	Hot wet summer + warm dry winter	'Wet-and-dry' tropical climate
Rainfall	Summer-concentrated, ~75-150 cm	Unreliable -> drought hazard
Vegetation	Tall grass + scattered trees	Parkland; grass shortens poleward
Trees	Drought-resistant, deciduous	Acacia, baobab; umbrella crowns

■ STATIC THEORY

- **Vegetation - 'parkland':** tall, coarse grasses (elephant grass up to ~5 m) with scattered, flat-topped, drought-resistant trees (acacia, baobab). Grass height and luxuriance DECREASE away from the equator (towards the desert margin) as rainfall falls.
- **Adaptations:** trees are deciduous (shed leaves in the dry season), have thick bark, long roots, umbrella crowns; grasses grow in tufts and turn yellow/dormant in the dry season. Annual fires (natural + human) are characteristic and maintain the grassland.
- **Fauna:** the savanna is the 'big-game' land - grazing herbivores (zebra, giraffe, antelope, wildebeest, buffalo, elephant) and predators (lion, cheetah); the natural home of large herds and of cattle.
- **Human life (GC Leong):** pastoralists like the Masai of East Africa (cattle herders) and cultivators like the Hausa of northern Nigeria (sorghum/millet growers) illustrate savanna livelihoods.
- **Farming hazards:** rainfall is unreliable, droughts long and severe; without irrigation, crop failures are common. The wet-dry regime causes rapid leaching/laterite formation, degrading soils.

■ MUST-KNOW FACTS

1. Koppen Aw; TRANSITIONAL climate between equatorial forest and hot desert.

■ UPSC TRAPS

✗ Savanna rainfall is spread evenly through the year.

2. Best developed in the Sudan belt (hence 'Sudan Climate'); also Llanos (Venezuela), Campo Cerrado (Brazil). 3. Two seasons: hot wet summer + warm dry winter; rain summer-concentrated (~75-150 cm). 4. Grass height DECREASES away from the equator; trees deciduous & drought-resistant (acacia, baobab). 5. The savanna is the natural home of grazing herds and cattle ('big-game country'); annual fires typical.	✓ It is strongly SEASONAL - concentrated in the hot summer; the winter is dry (wet-and-dry climate). ✗ Savanna grass is uniform everywhere. ✓ Grass is tall & lush near the equatorial margin and becomes short & sparse towards the desert margin as rain declines. ✗ Savanna is a purely natural climax vegetation untouched by humans. ✓ Human-set annual fires strongly shape and maintain the savanna grassland (fire-climax).
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■ MAINS ANGLE

Explain why the savanna is called a 'transitional' climate, and how its wet-dry rhythm shapes vegetation, wildlife and the precariousness of savanna agriculture.

■ STUDY LINK

Geography -> World Climatic Types -> Savanna (Aw); links to tropical grassland biome & desertification

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: Africa's Great Green Wall & the Sahel Greening Debate

GS-I (World Geography) +
GS-III
(Environment / Land Degradation)
Geography -
Current Affairs

■ NEWS TRIGGER

2024-26: the African Union's Great Green Wall (GGW) - a 8,000 km, 21-country belt from Senegal to Djibouti - remains ~20% complete against its 2030 target, hampered by funding gaps, conflict and drought; meanwhile studies debate genuine 'Sahel greening' from better rains and land management.

■ INTRO & ORIGIN

The Sahel is the real savanna-desert transition zone, and the Great Green Wall is the world's biggest attempt to halt desertification along it - the applied case study for the savanna chapter's 'farming hazard / land degradation' theme.

GGW launched 2007 by the African Union to restore ~100 million ha of degraded land by 2030, sequester ~250 million t carbon and create ~10 million jobs across the Sahel.

■ KEY DATA

Aspect	Detail	Status
Launch	African Union, 2007	8,000 km, Senegal -> Djibouti
Target 2030	Restore 100 mn ha, 250 mn t C, 10 mn jobs	~20% achieved by 2024/25
Barriers	Funding gap, conflict, drought, locusts	Slow progress
Sahel greening	Localised vegetation gains	Debated; net degradation persists

■ STATIC THEORY

- **Desertification:** land degradation in arid/semi-arid/dry sub-humid areas from climate and human pressure; the Sahel is its global poster-child (UNCCD focus).
- **Great Green Wall = nature-based solution:** mosaic restoration (not a literal single line of trees) using native species, water harvesting and community land management for adaptation + mitigation.
- **UN Decade on Ecosystem Restoration (2021-2030):** the wider framework channelling finance and attention to GGW-type restoration.

■ MUST-KNOW FACTS

1. Great Green Wall: African Union, launched 2007; ~8,000 km across 21 Sahel countries (Senegal-Djibouti).
2. Target by 2030: restore ~100 mn ha, sequester ~250 mn t carbon, create ~10 mn jobs.
3. Only ~20% complete by 2024/25; funding gaps, conflict and drought slow it.
4. The Sahel is the classic savanna/desert transition; UNCCD leads the anti-desertification framework.
5. GGW is now understood as a mosaic of restored ecosystems, not a single tree-line.

■ UPSC TRAPS

- ✗ The Great Green Wall is a continuous single row of trees across Africa.
- ✓ It is a MOSAIC of restored land uses (agroforestry, water harvesting, native regeneration), not a literal single line of trees.
- ✗ The Sahel is unambiguously greening and desertification is solved.
- ✓ Localised greening exists, but net land degradation and drought risk persist; the debate is unsettled.

■ MAINS ANGLE

Assess the Great Green Wall as a nature-based response to Sahel desertification - its ambitions, the reasons for slow progress, and lessons for dryland restoration worldwide.

■ STUDY LINK

Geography -> Savanna/Sahel + GS-III desertification (UNCCD, ecosystem restoration); contrast with India's Aravalli Green Wall (Advanced tier)